Student 26 **MONISH.S** 192311132RD 2 / 2 submitted

Q1. Smart Hospital Patient Registration System

CODE

```
import java.util.*;
class Main{
    static ArrayList<Patient> list=new ArrayList<>();
    static HashMap<String,Patient> map=new HashMap<>();
    static String properCase(String s){
        String[] words=s.toLowerCase().trim().split("\\s+");
        String ans="";
        for(String w:words){
            ans+=Character.toUpperCase(w.charAt(0))+w.substring(1)+" ";
        }
        return ans.trim();
    }
    static void registerPatient(Scanner sc){
        Patient p=new Patient(sc.nextLine(),properCase(sc.nextLine()),sc.nextLine(),sc.nextLine().toUpperCase(),sc.nextLine());
        list.add(p);
        map.put(p.id,p);
        System.out.println("Patient Registered Successfully.");
    }
    static void searchPatient(Scanner sc){
        String id=sc.nextLine();
        if(map.containsKey(id)){
            Patient p=map.get(id);
            System.out.println(p.id+" "+p.name+" "+p.mobile+" "+p.department+" "+p.doctor);
        }
        else{
            System.out.println("Patient Not found");
        }
    }
    static void displayPatients(){
        for(Patient p:list)
            System.out.println(p.id+" "+p.name+" "+p.department);
    }
    static void department(){
        HashMap<String,Integer> hm=new HashMap<>();
        for(Patient p:list)
            hm.put(p.department,hm.getOrDefault(p.department,0)+1);
        for(String s:hm.keySet())
            System.out.println(s+":"+hm.get(s));
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        while(true){
            int choice=Integer.parseInt(sc.nextLine());
            switch(choice){
                case 1:
                    registerPatient(sc);
                    break;
                case 2:
                    searchPatient(sc);
                    break;
                case 3:
                    displayPatients();
                    break;
                case 4:
                    department();
                    break;
                case 5:
                    return;
                default:
                    System.out.println("Invalid Choice");
            }
        }
```

```
    }
}
class Patient{
    String id,name,mobile,department,doctor;

    Patient(String id,String name,String mobile,String department,String doctor){
        this.id=id;
        this.name=name;
        this.mobile=mobile;
        this.department=department;
        this.doctor=doctor;
    }
}
```

INPUT

```
1
P101
Rahul kumar
9876543210
Cardiology
Dr.Arun
5
```

OUTPUT

```
(no output)
```

Q2. Online Library Book Management System

CODE

```
import java.util.*;
class Main{
    static ArrayList<Book> list=new ArrayList<>();
    static HashMap<String,Book> map=new HashMap<>();
    static HashSet<String> authors=new HashSet<>();

    static String titleCase(String s){
        String[] a=s.trim().toLowerCase().split(" ");
        String ans="";
        for(String i:a){
            ans+=Character.toUpperCase(i.charAt(0))+i.substring(1)+" ";
        }
        return ans.trim();
    }
    static void addBook(Scanner sc){
        String id=sc.nextLine().trim();
        if(map.containsKey(id)){
            System.out.println("Duplicate Book ID");
            return;
        }
        String title=titleCase(sc.nextLine());
        String author=sc.nextLine().trim();
        String category=sc.nextLine().trim().toUpperCase();
        String publisher=sc.nextLine().trim();
        Book b=new Book(id,title,author,category,publisher);
        list.add(b);
        map.put(id,b);
        authors.add(author);
        System.out.println("Book Added Successfully.");
    }
    static void searchBook(Scanner sc){
        String id=sc.nextLine();
        if(map.containsKey(id)){
            Book b=map.get(id);
            System.out.println("Book ID: "+b.id);
            System.out.println("Title: "+b.title);
            System.out.println("Author: "+b.author);
            System.out.println("Category: "+b.category);
            System.out.println("Publisher: "+b.publisher);
        }
        else{
            System.out.println("Book Not Available");
        }
    }
    static void displayBooks(){
        System.out.println("ID TITLE AUTHOR CATEGORY");
        for(Book b:list){
            System.out.println(b.id+" "+b.title+" "+b.author+" "+b.category);
        }
    }
    static void categoryCount(){
        HashMap<String,Integer> count=new HashMap<>();
        for(Book b:list){
            if(count.containsKey(b.category))
                count.put(b.category,count.get(b.category)+1);
            else
                count.put(b.category,1);
        }
        for(String s:count.keySet()){
            System.out.println(s+": "+count.get(s));
        }
    }
```

```
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        while(true){
            int ch=Integer.parseInt(sc.nextLine());
            switch(ch){
                case 1:
                    addBook(sc);
                    break;
                case 2:
                    searchBook(sc);
                    break;
                case 3:
                    displayBooks();
                    break;
                case 4:
                   categoryCount();
                    break;
                case 5:
                    return;
                default:
                    System.out.println("Invalid Choice");
            }
        }
    }
}
class Book{
    String id,title,author,category,publisher;
    Book(String id,String title,String author,String category,String publisher){
        this.id=id;
        this.title=title;
        this.author=author;
        this.category=category;
        this.publisher=publisher;
    }
}
```

INPUT

```
1
B101
Java Programming
Herbert Schildt
Programming
McGraw Hill
3
5
2
B101
McGraw Hill
6
```

OUTPUT

```
(no output)
```